Roll Number: _____ Name_____ Group_____

THAPAR INSTITUTE OF ENGINEERING &TECHNOLOGY, PATIALA
Department of Computer Science & Engineering
Database Management System (UCS310) – End Semester Test
Date: 03/06/2025 & Time: 2:00PM ***MM: 40 & MT: 180 Min***

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***Note:** Attempt any five questions with all sub-parts like (a), (b) for each question **in one place.** **Do mention Page No. of your attempt on the front page of the answer sheet.** Assume missing data (if any). Show all intermediate computations properly.*

Q. No.		Question	Marks	CO	BL
Q1	(a)	Consider the following two transactions running concurrently. i) Assume A=1000 in S1, if these transactions execute concurrently without any concurrency control, demonstrate the **lost update** and **dirty read** problem separately with a step-by-step interleaving of transaction instructions in schedule S1 and show the final value of A. ii) Apply Strict Two-Phase Locking (2PL) protocol on transactions given in schedule S2. If initial value of A and B is 0 find out final values of A and B after execution of T1 and T2. **S1** T1 | T2 READ (A); | READ (A); A := A - 50; | A := A + 100; WRITE (A); | WRITE (A); **S2** T1 | T2 READ(A) | READ(B) READ(B) | READ(A) IF A=0 Then B=B+1 | IF B!=0 Then A=A-1 WRITE(B) | WRITE(A)	2 2	4	3 3
	(b)	Consider the following database schedule involving two transactions, T_1 and T_2: $S = R_2(X); R_1(X); R_2(Y); W_1(X); R_1(Y); W_2(X); C_2; A_1$ Ri, Wi, Ci and Ai means Read, Write, Commit and Abort operation of transaction Ti, determine with reasoning whether the schedule S is recoverable, irrecoverable, cascading, or cascadeless.	2	4	3
	(c)	Consider the relation ***Orders (order_id, customer_name, order_date, order_amount, order_status)***. Write a PL/SQL block that uses **parameterized cursor** to display all orders with a specific status (e.g., 'Pending', 'Shipped', Delivered). The cursor should accept the order status as a parameter, and for each order matching that status, the block should print the ***order_id, customer_name, and order_amount.***	2	5	3
Q2	(a)	A university's online registration system allows students to enroll in courses and deducts the course fee from their accounts during registration. During peak registration time, two students attempted to register for the last available seat in Course X at the same moment. Both transactions read the seat availability as '1' and were allowed to proceed. Student A's registration went through, and their fee was deducted. Student B's registration process failed midway due to a system crash — their seat was still marked as occupied, but the fee was not deducted. The course now shows 2 students registered for 1 seat, and Student B is enrolled without payment. Prepare a tabular representation with the following four columns: (i) ACID Property violated in the given scenario, (ii) Component of DBMS Responsible to handle violated properties, and (iii) Why These Properties are important to hold, (iv) Explain how these violations occurred in the given scenario.	4	4	4
	(b)	Consider the three transactions T1, T2, and T3, and the schedules S1 and S2 given below. Draw the precedence graphs for S1 and S2, and state whether each schedule is serializable (conflict and view) or not. If a schedule is serializable, write down the equivalent serial schedule(s). **T1: R1 (X); R1 (Z); W1 (X); T2: R2 (Z); R2 (Y); W2 (Z); W2 (Y); T3: R3 (X); R3 (Y); W3 (Y);** S1: R1 (X); R2 (Z); R1 (Z); R3 (X); R3 (Y); W1 (X); W3 (Y); R2 (Y); W2 (Z); W2 (Y); S2: R1 (X); R2 (Z); R3 (X); R1 (Z); R2 (Y); R3 (Y); W1 (X); W2 (Z); W3 (Y); W2 (Y);	4	4	5
Q3	(a)	i) Prepare a tabular representation with the following three columns: **Column1:** Normal Form (2NF to 5NF each in 1 row), **Column2:** Requirement to satisfy the Normal Form, Column**3:** Explanation of the Dependency specified in column2.	2	3	5

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		ii) Let R (A, B, C, D, E, P, G) be a relational schema in which the following functional dependencies are known to hold: AB→CD, DE→P, C→E, P→C and B→G. Find out the highest normal form of relation.	2		
	(b)	Explain the difference between a subquery and a correlated subquery in SQL. Consider a database with two relations: ***Employees(emp_id, emp_name, department_id, salary), Departments(department_id, department_name).*** Write a query to: i) Find the names of employees who work in the 'Sales' department using subquery. ii) Find the names of employees whose salary is greater than the average salary in their own department using correlated subquery.	4	5	3
Q4	(a)	Refer relation ***employees (emp_id, emp_name, department_id, salary)*** i) Write a PL/SQL block to find and increase the salary of the employee with the highest salary in department 20 by 5%. ii) Create a PL/SQL function adjust_salary_if_needed(emp_id) that checks an employee's salary from the employees relation. If it's below 25,000, increase it by 15% and return 'Salary increased by 15%'; otherwise, return 'No increment needed'.	2 2	5	3
	(b)	Consider the following four variants of Two-Phase Locking (2PL): Simple 2PL, Conservative (Pre-Claiming) 2PL, Strict 2PL, and Rigorous 2PL. Compare these four protocols based on the following criteria: ***a) Possibility of Deadlock, b) Cascadelessness, c) Level of Concurrency, d) Possibility of Dirty Read problem.*** Construct a comparative table having protocols names in columns and criteria in rows and briefly justify your entries in 1 line per row.	4	4	5
Q5	(a)	Compare four common database recovery techniques based on their use and Briefly discuss their impact on performance and reliability. Consider the following log sequence of two transactions on a bank account, with initial balance 12000, that transfer 2000 to a mortgage payment and then apply a 5% interest. Suppose the database system crashes just before log record 7 is executed. When the system is restarted, which log records must go into redo list and undo list. <Checkpoint> Log record 1. T1 start Log record 2. T1 B old=12000 new=10000 Log record 3. T1 M old=0 new=2000 Log record 4. T1 commit Log record 5. T2 start Log record 6. T2 B old=10000 new=10500 ---CRASH--- Log record 7. T2 commit	4	4	5
	(b)	Create a trigger named **marks_update_log** that activates after update of marks on the relation ***Student (Stud_id, Name, marks, cgpa).*** The trigger should log the following details into a relation named ***Marks_Log(stud_id, old_marks, new_marks, update_date).***	4	5	6
Q6	(a)	i) Draw the Transaction State Diagram and describe each state and transition involved. ii) Consider the relation scheme R (A, B, C, D, E, H), ***FDs: A → B, BC → D, E → C, D → A,*** what are the candidate keys of R and write prime and non-prime attributes?	2 2	4 2	3
	(b)	A relational database contains two relations Student and Performance as shown below: ***Student (Roll No, Student_Name), Performance (Roll No, Subject_Code, Marks).*** Write output for following SQL queries Table: student Roll_no | Student_name 1 | Amit 2 | Priya 3 | Vinit 4 | Rohan 5 | Smita Table: Performance Roll_no | Subject_code | Marks 1 | A | 86 1 | B | 95 1 | C | 90 2 | A | 89 2 | C | 92 3 | C | 80 1. SELECT S.Student_name, sum(P.Marks) FROM Student S Natural Join Performance P WHERE P.Marks >84 GROUP BY S.Student_name. 2. SELECT s.Student_name FROM Student s LEFT JOIN Performance p ON s.Roll_no = p.Roll_no WHERE p.Roll_no IS NULL; 3. SELECT Subject_code FROM Performance GROUP BY Subject_code HAVING AVG(Marks) = (SELECT MAX(AVG(Marks)) FROM Performance GROUP BY Subject_code); 4. SELECT COUNT(*) FROM (SELECT Roll_no FROM Performance GROUP BY Roll_no HAVING COUNT(DISTINCT Subject_code) >= 2);	4	5	3